

Auger (Screw) Filler — Equations, Justification and References

This document lists all mathematical equations used for the auger filler and related systems, with explanations, justifications and direct links to the academic/industry sources used. Each equation is shown in a clear mathematical form, followed by the source(s) and reason it applies to our design (salt powder, D=60 mm, etc.).

1. Basic density relation

Equation: $\rho = m / V \Rightarrow V = m / \rho$

Justification: Fundamental definition of density; used to convert target bag mass to required volumetric fill. Applies universally to materials and is the starting point for volumetric dosing calculations.

Sources: OpenStax College Physics; Wikipedia. Links:
<https://openstax.org/books/college-physics-2e/pages/11-2-density> ;
<https://en.wikipedia.org/wiki/Density>

2. Auger / Screw volumetric relations (per revolution)

Equation (basic geometric volume per revolution): $V_{rev} \approx A \times p \times FF$ where $A = \pi \cdot D^2 / 4$ (if using outer diameter), p = pitch, FF = volumetric fill factor

Alternative (more accurate accounting for shaft core): $V_t = (\pi/4) \cdot (D^2 - D_s^2) \cdot p$

Justification: The auger transports material via the helical cavity formed between flights and shaft; the theoretical cavity volume per revolution equals the helix axial volume (area $\times$ pitch). The fill factor $FF < 1$ accounts for powder not fully occupying the cavity due to bridging, segregation and packing. These relations are standard in screw feeder sizing literature.

Sources: Bortolamasi & Fottner — Design and Sizing of Screw Feeders (PDF). Link: https://mediatum.ub.tum.de/doc/1188322/fml_20131230_514_export.pdf ; RWTH thesis 'Machine Design for Auger Dosing of Powders' <https://publications.rwth-aachen.de/record/814145/files/814145.pdf>

3. Revolutions per fill and RPM

Equations: $revs = V_{bag} / V_{rev}$; $RPM = (revs \times 60) / t_{fill}$

Justification: Number of revolutions required is target volume divided by volume per revolution. RPM ties revolutions to time for servo sizing and control.

Source: Bortolamasi & Fottner (see link above)

4. Mechanical power, torque and shaft sizing

Equations: $\omega = 2\pi \cdot \text{RPM}/60$ (rad/s) ; $P = T \cdot \omega \Rightarrow T = P/\omega$; $\tau_{\text{max}} = 16 \cdot T / (\pi \cdot d^3) \Rightarrow d = (16 \cdot T / (\pi \cdot \tau_{\text{allow}}))^{1/3}$

Justification: Standard rotational dynamics and torsion relations used to size motor and shaft. $P = T \cdot \omega$ links mechanical power to torque and angular speed; torsion formula for a solid circular shaft gives the maximum shear stress for a given torque.

Sources: Roark's Formulas for Stress and Strain (torsion tables) and Wikipedia 'Torsion (mechanics)': [https://en.wikipedia.org/wiki/Torsion_\(mechanics\)](https://en.wikipedia.org/wiki/Torsion_(mechanics)) ; Roark's reference (general).

5. Volumetric fill factor (FF) — empirical considerations

Explanation: FF represents the fraction of the theoretical cavity volume actually filled by bulk powder per revolution. FF depends on material bulk density, particle size, shape, flowability, and auger geometry (clearance, helix angle, pitch). FF is typically derived from experimental tests or manufacturer tables; common starting values for free-flowing powders range 0.4–0.8.

Sources: Bortolamasi & Fottner; KWS Screw Conveyor guide; research on predictive models:

https://www.researchgate.net/publication/329993867_A_multivariate_approach_to_predict_the_volumetric_and_gravimetric_feeding_behavior_of_a_low_feed_rate_feeder_based_on_raw_material_properties

6. Hopper flow, bridging and Jenike methodology

Relevance: Proper hopper geometry (cone angle, outlet diameter) and flow regime (mass flow vs funnel flow) are essential to supply the auger with consistent feed and avoid bridging or rat-holing. Use Jenike analysis or flow testing for final hopper design.

Sources and reading: Jenike & Johanson Hopper Design principles (summaries), Funnel flow studies and practical solids handling slides:

<https://www.scribd.com/document/744031259/Hopper-Design-Principles-Jenike-Johanson> ;

https://www.researchgate.net/publication/30759276_Funnel_Flow_in_Hoppers ;

https://static.horiba.com/fileadmin/Horiba/Products/Scientific/Particle_Characterization/Webinars/Slides/Practical_Advice_for_Storing_and_Handling_Bulk_Solids_Webinar.pdf

7. Heat sealing OPP/PP film (sealing parameters)

Relevance: Seal temperature, dwell time and pressure determine seal integrity for OPP/PP laminates used as packaging film. Test and determine operating point (temperature/pressure/time) for your specific film roll.

Key references: Tetsuya Tsujii et al., 'Effect of heat-sealing temperature on the properties of OPP/PP heat seals' (J. Appl. Polym. Sci.). Link:

https://www.researchgate.net/publication/230207662_Effect_of_heat-sealing_temperature_on_the_properties_of_OPPCPP_heat_seals_Part_II_Crystallinity_and_thermomechanical_properties ; UTeM study on bar sealing parameters: https://eprints.utm.edu.my/id/eprint/10296/1/EPL-0000404_article-utm.pdf

8. Load cell, HX711 and weighing accuracy

Relevance: Load cell (with amplifier HX711) provides feedback for gravimetric trimming or to calibrate the volumetric auger. Proper mounting and calibration are crucial for accuracy.

Key reference: Rice Lake Load Cell and Weigh Module Handbook (practical guide). Link: https://www.ricelake.com/media/b5mlti1q/m_22054_lc-wm_handbook_enus_rev1-1.pdf

9. Studies on dosing accuracy and control

References on improving dosing accuracy, adaptive models and controls for auger fillers: F. Kruppa 'Increasing the dosing accuracy of a screw dosing device' (2023). Link: <https://onlinelibrary.wiley.com/doi/full/10.1002/pts.2703> ; Adaptive modeling (ASABE). Link: <https://elibrary.asabe.org/azdez.asp?AID=38131&CID=loui2011&JID=5&T=1>

10. Reference list (links)

- Design and Sizing of Screw Feeders — M. Bortolamasi & J. Fottner (PDF): https://mediatum.ub.tum.de/doc/1188322/fml_20131230_514_export.pdf
- Machine Design for Auger Dosing of Powders — RWTH thesis (PDF): <https://publications.rwth-aachen.de/record/814145/files/814145.pdf>
- Jenike & Johanson — Hopper Design Principles (summary): <https://www.scribd.com/document/744031259/Hopper-Design-Principles-Jenike-Johanson>
- Funnel flow in Hoppers (research): https://www.researchgate.net/publication/30759276_Funnel_Flow_in_Hoppers
- Effect of heat-sealing temperature on OPP/CPP heat seals — Tetsuya et al. (ResearchGate): https://www.researchgate.net/publication/230207662_Effect_of_heat-sealing_temperature_on_the_properties_of_OPPCPP_heat_seals_Part_II_Crystallinity_and_thermomechanical_properties
- Rice Lake Load Cell and Weigh Module Handbook (PDF): https://www.ricelake.com/media/b5mlti1q/m_22054_lc-wm_handbook_enus_rev1-1.pdf
- KWS Screw Conveyor Guide (capacity): <https://www.kwsmfg.com/engineering-guides/screw-conveyor/screw-conveyor-capacity/>

- Increasing dosing accuracy of a screw dosing device — F. Kruppa (2023):
<https://onlinelibrary.wiley.com/doi/full/10.1002/pts.2703>
- Adaptive Modeling for Weight Control in Vertical Auger Fillers (ASABE):
<https://elibrary.asabe.org/azdez.asp?AID=38131&CID=loui2011&JID=5&T=1>